

Chapter 123 — Surgical Procedures for Sphincteric Incontinence in the Male: The Artificial Urinary Sphincter and Perineal Sling Procedures

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- **Male sphincteric urinary incontinence (SUI)** significantly impairs **quality of life (QoL)**; causes include congenital anomalies, injury and genitourinary surgery.
- **Radical prostatectomy (RP) for prostate cancer (CaP) is a leading cause: storage fails because external sphincter resistance is insufficient.**
- **US burden is high** because RP is common. **It may decline** with better technique, fewer RPs after the 2012 U.S. Preventive Services Task Force (USPSTF) recommendation against prostate-specific antigen (PSA) screening, and more active surveillance.
- **QoL impact drives treatment-seeking**; slings and the artificial urinary sphincter (AUS) **increase outlet resistance**. Physician and patient perceptions are barriers. **Decisional regret is highest after conservative treatment**, so outreach is needed.
- **AUS = gold standard**, because it treats **all degrees** of SUI. **Bulbar slings** suit **mild-to-moderate** urinary incontinence (UI), with long-term data now available.

CLASSIFICATION, PATHOPHYSIOLOGY AND AETIOLOGY

- **Continence needs** bladder, **internal sphincter** (bladder neck, prostatic urethra) and **external sphincter** (rhabdosphincter). **Male SUI requires both sphincters to be impaired.**

Internal sphincter incompetence	External sphincter incompetence
Pelvic surgery or radiation; bladder neck injury; specific sympathetic neuropathic dysfunction; embryological disruption	Commonest: after RP. Also pelvic radiation; pelvic fracture urethral injury; traumatic or acquired myelopathy; congenital — spinal dysraphism, sacral agenesis, exstrophy-epispadias complex

- **Reduced compliance, detrusor overactivity (DO) and urgency urinary incontinence (UUI)** complicate diagnosis — evaluate carefully.
- **Prevalence rises with age: 11% at 60–64, 31% at >85**; incidence tracks the causative procedures.
- **Post-RP rates vary widely** (patient, surgeon, data method); **self-report gives higher rates** than chart or physician record. **Continence improves for up to 2 years.**
- **Nerve sparing** (preserving pudendal innervation of the rhabdosphincter) gives **faster, not better final, continence** (meta-analyses). **Autonomic afferent denervation and reduced membranous urethral sensitivity** are associated with SUI.
- **Other mechanisms:** rhabdosphincter, pelvic floor and bladder neck damage; incontinent men differ in **urethral pressure profile, length and sphincter morphology**. **Posterior reconstruction** speeds recovery, **not long-term continence.**
- **Bladder neck stenosis** lowers continence, mainly via secondary procedures damaging the sphincter. **Vesicourethral anastomotic stenosis (VUAS)** may impair coaptation, and **its treatment can cause SUI** when the contracture reaches the membranous urethra (endoscopically: lost coaptation, radiating folds).

- **Climacturia** (leakage with arousal or orgasm): little studied; neurogenic mechanisms proposed.

Concept — Male *SUI* needs **failure of both sphincters**. *RP* removes the internal sphincter by design, so after *RP* continence rests entirely on the rhabdosphincter, and any insult to it — nerve, muscle, stricture treatment — is exposed.

MECHANISM OF CONTINENCE WITH SURGICAL DEVICES

Compression Devices

- **AUS (AMS 800)**: fluid-filled cuff giving **circumferential compression about 2 cm long**. The **pressure-regulating balloon (PRB)** sets the compression, chosen for tissue and cuff site. **Standard bulbar PRB 61–70 cm H₂O** balances occlusion against erosion; **lower pressure** trades continence for less erosion.
- **An open cuff lowers resistance**, so men without useful detrusor contraction void by straining or **Credé manoeuvre**.
- **ProACT (Adjustable Continence Therapy)**: two silicone balloons beside the bladder neck, filled with saline through **scrotal ports**.

Fig. 123.1 — AMS 800: cuff, pump and PRB. **The 3.5-cm cuff has a unique three-cushion design.**

Male Slings

Nonadjustable Slings

- **Transobturator slings reposition a residual rhabdosphincter**: leak point pressure rises without obstruction; fluoroscopy and magnetic resonance imaging (MRI) show **urethral elongation** and elevation of bladder neck, posterior wall and rhabdosphincter.
- **Virtue**: four-armed mesh; prepubic arms compress against the urogenital diaphragm.

Adjustable Slings

- **ATOMS**: central cushion under the bulbar urethra, titanium–silicone scrotal port for saline adjustment.

Fig. 123.2 — Proposed sling mechanisms: (A) transobturator; (B) quadratic fixation.

DEVICE MECHANISMS COMPARED

Device	Mechanism	Site	Adjustable	Needs functioning sphincter?
AUS (AMS 800)	Circumferential compression, patient-cycled	Bulbar urethra or bladder neck	No (PRB pressure chosen at surgery)	No
ProACT	Lateral balloon compression	Bladder neck	Yes — scrotal ports	—
Advance / Advance XP	Repositioning and lengthening, no compression	Bulbar urethra, transobturator	No	Yes
Virtue	Compression (prepubic) plus elevation (transobturator)	Bulbar urethra, four arms	No	Less dependent (compressive)
ATOMS	Cushion compression of bulbar urethra	Bulbar urethra, transobturator	Yes — scrotal port	Yes

- **What the table tests**: the key division is **repositioning (Advance) versus compression (all others)**. Only a repositioning sling depends absolutely on a coapting residual sphincter. **Trap**: ATOMS and Virtue are "slings" but act by compression.

Concept — Every device either **compresses the urethra** or **repositions a functioning sphincter**. Which one the patient needs is decided by how much sphincter is left.

EVALUATION AND DIAGNOSIS

- **Core:** history, examination, **postvoid residual (PVR)**, urinalysis; **cystoscopy and pressure-flow urodynamics (UDS)** as indicated for VUAS, residual sphincter and storage.

History

- **Focus:** UI type and severity; prior surgery, radiotherapy, radiation cystitis; neurological symptoms.
- **Most important step: classify the UI** — SUI (stress or activity-related), UUI, mixed or overflow.
- **Incontinence during sleep and nocturia** help separate **severe SUI** from urgency.
- **In men treated for CaP, establish** disease status, the impact of planned adjuvant therapy, and life expectancy.

Physical Examination

- **Examine** abdomen, back, genitalia, perineum, rectum and nervous system. **Assess manual dexterity if AUS is considered.**
- **Treat skin breakdown or infection before surgery. Note incisions** for PRB planning, and **inguinal hernia or repair** if the external-ring route is planned. **Scrotal pathology** (hydrocele, hernia, mass) affects **pump** placement.

Patient Reported Outcome Measures

- **Severity and bother guide the operation;** patient-reported outcome measures (PROMs) add to history and examination.
- **UCLA Prostate Cancer Index and Expanded Prostate Cancer Index Composite (EPIC)** capture post-RP SUI and UUI, mainly in research; clinical performance and responsiveness are **not formally assessed**. **Incontinence Symptom Index and Incontinence Impact Questionnaire-7** have measured AUS outcomes but were **not developed for male SUI**.

Voiding Diaries

- **2–7 days:** UI episode frequency, **bladder capacity**, and **urgency or UUI** (separates SUI from UUI).

Pad Weight Testing

- **Objective severity measure**, since pad counts and change thresholds vary: pre-weighed pads worn (usually **24 hours**) → reweighed → difference = urine loss.
- **Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction (SUFU) Pad Test:** questionnaire on number, size and wetness of 24-hour pads — **correlates with actual loss, not validated to predict surgical outcome**.

Standing Cough Test

- **Office test** (stand, cough forcefully, observe); **correlates with 24-hour pad weight**, which rises with grade. Office Valsalva also demonstrates SUI.

STANDING COUGH TEST GRADING

Grade	Finding
0	Leakage reported but not demonstrable
1	Delayed drops only
2	Early drops , no stream
3	Delayed stream
4	Early and persistent stream

- **Pattern:** grade rises with **timing (delayed → early)** and **form (drops → stream)**. Trap: grade 0 is a reported but *undemonstrated* leak, not a negative history.

Laboratory

- **Urinalysis and culture mandatory;** investigate abnormalities; **eradicate urinary tract infection (UTI).**
- **Creatinine and PSA** (renal function, cancer status); **renal insufficiency needs evaluation first.**

Cystoscopy

- **Mandatory if VUAS, bladder neck contracture or stricture is suspected** — unrecognised obstruction worsens SUI surgery.
- **Grades residual sphincter function to select for a transobturator sling; poorly coapting sphincter → AUS.**
- Also used **after AUS** (mechanical failure, leak, atrophy) and **after explant for erosion**, before reimplantation, to exclude **stricture or diverticulum**.

Fig. 123.4 — External sphincter after RP with a **quadrant defect from 9 to 12 o'clock** and incomplete circumferential coaptation. Such a man is a **poor transobturator sling candidate**, because that sling relies on repositioning a functional sphincter.

Urodynamics

- **Capacity, compliance, contractility;** intrinsic sphincter deficiency in nearly all.
- **Not routine; use when diagnosis is in doubt: mixed UI, SUI not shown in clinic, overflow, suspected poor compliance.**
- **Mixed UI with DO:** counsel that **UUI may persist.**
- **Poor compliance is the more serious finding** (high-pressure storage harms kidneys). It **does not predict worse AUS continence** — so good continence may hide **silent upper-tract deterioration.**
- **Detrusor hypocontractility → AUS:** the bladder may not overcome a sling's added resistance.

FIG. 123.3 AUA/SUFU GUIDELINE EVALUATION ALGORITHM FOR INCONTINENCE AFTER PROSTATE TREATMENT

From Sandhu JS, Breyer B, Comiter C, et al. Incontinence after prostate treatment: AUA/SUFU Guideline. J Urol. 2019;202(2):369–378.

Initial presentation (history, examination, relevant tests) → three branches:

- **UUI or urgency-predominant mixed UI (MUI)** → American Urological Association (AUA)/SUFU overactive bladder pathway.
- **Other pathology suspected** (microhaematuria, low flow, high PVR) → **cystoscopy and/or UDS** → treat stricture (interval cystoscopy) or other pathology → rejoin SUI arm.
- **SUI or stress-predominant MUI** → **pelvic floor exercises or therapy (PFME/PFMT)**, discuss goals → follow-up. Improved, not bothered → stop. Otherwise → **discuss surgical and non-surgical options** → surgical algorithm, non-surgical care or none; **surgery stays open later.**

KEY POINTS — EVALUATION

- History and examination characterise type, severity and capacity in most.
- **Uroflowmetry and PVR** screen for obstruction.
- Obstruction, overactivity or poor contractility → **UDS**; urethral, vesicourethral or bladder neck pathology → **cystoscopy**; **urinalysis ± culture** before surgery.

Concept — Evaluation asks **three questions before any device**: is this really SUI, is the bladder safe to store against more resistance, and how much sphincter remains? The answers select the device.

INDICATIONS FOR SURGERY

- **Indication: irreversible sphincter incompetence with UI impairing QoL.**
- **Pelvic floor muscle training (PFMT)** ± biofeedback or electrical stimulation works after RP: **shortens time to continence**, improves 1- and 3-month continence, **no benefit beyond 12 months**.
- **Do not delay surgery for severe or gravitational SUI unimproved at 6 months**, especially with a cystoscopic sphincter defect.
- **Shared decision-making**; weigh QoL gain against **performance status and life expectancy**.
- **First-line: AUS and perineal slings. Bulking agents are outside the post-RP and post-transurethral resection of the prostate (TURP) algorithm** — only for **small-volume leakage in men unfit for more**.
- **Choice also depends on** cognition, dexterity, body mass index, prior radiotherapy, prior surgery including urethroplasty, and bladder function (e.g. neurogenic).

Absolute — alternative to SUI surgery required	Relative contraindications
Low-compliance bladder threatening renal function (vesicoureteric reflux at low pressure) Low-capacity bladder (would cause severe frequency or UII) Tissue at bladder neck or urethra too poor for a sling or AUS → clamps, catheter, bladder neck closure or urinary diversion	Frequent future transurethral management needed — bladder cancer, refractory VUAS, radiation cystitis. Repeated instrumentation raises the risk of erosion and infection

- **Radiation cystitis** is hardest (large-bore catheters for clots): **hyperbaric oxygen and 6 months free of refractory bleeding before AUS**; **repeated clot retention** → cystectomy and diversion.

Compression Devices**Artificial Urinary Sphincter**

- **Standard for moderate-to-severe SUI** with **adequate cognition and dexterity**; works **without detrusor contraction** and **after radiation**. AMS 800 has the **longest follow-up**; social continence = **0–1 pad/day** (outcomes in table below).
- **Success predictors**: surgeon experience, volume, intact compliance. **Reoperation is common**.

ProACT

- Fewer pads, better QoL. **Versus male sling (randomised)**: **similar continence, more erosion; worse than sling after radiotherapy**.

Slings

- **Perineal slings** — **mild-to-moderate SUI**, nonadjustable or adjustable. **Commonest: Advance / Advance XP (transobturator)**. **Adjustable slings have limited data**; **ATOMS beats ProACT** on dryness, improvement, satisfaction and pad use.

- **Sling failure: radiation, severity, previous surgery.**
- **Severity thresholds (24-hour pad weight): mild <150 g; moderate <400 g.** Transobturator slings are primarily for mild-to-moderate UI.
- **MASTER trial** — randomised controlled trial (RCT) of AUS versus male sling. **Primary result: the sling was non-inferior to AUS for continence at 12 months.** The chapter's clinical emphasis is **AUS superiority where 24-hour pad weight is >250 g or after radiotherapy.**
- **Slings as an alternative in more severe SUI:** men who refuse AUS for fear of infection, erosion or mechanical failure, and men with **limited physical or cognitive capacity.**

DEVICE OUTCOMES COMPARED

Device	Target severity	Headline efficacy	Weak point
AUS	Moderate-severe; all degrees	Cure about 56%; social continence 55–77%; satisfaction about 90%	Revision 10–29% at 3 years
Advance XP	Mild-moderate	Cure about 60% (meta-analysis); durable (about 57% cured, 25% improved)	Poor after RT, severe UI, previous surgery
I-STP TOMS	Mild-moderate	Deteriorates: about 38% at 1 year → 15% at 5 years	Poor durability
ATOMS	Mild-moderate	About 67% dry; 90% improved; 16% complications	Limited data
ProACT	—	About 60% dry	More erosion; inferior after RT

- **Pattern: cure rates cluster around 55–65%** whichever device is used. The devices differ in **durability, erosion and eligibility**, not headline cure. Trap: these outcomes are not comparable head-to-head, because the populations differ in severity.

FIG. 123.5 AUA/SUFU ALGORITHM FOR SURGICAL MANAGEMENT

From Sandhu JS, Breyer B, Comiter C, et al. Incontinence after prostate treatment: AUA/SUFU Guideline. J Urol. 2019;202(2):369–378.

Assess **degree of incontinence** and **radiation history** (primary, adjuvant or salvage) and discuss goals →

Stratum	Recommendation	Options offered
Severe UI or any history of radiotherapy (RT)	AUS first-line; sling only with appropriate counselling	AUS; male sling
Moderate UI, no RT	Discuss AUS and male sling, possibly other devices	AUS; sling; other (bulking — rarely durable; adjustable balloons — specialised centres)
Mild UI, no RT	Discuss continued PFMT, male sling, possibly AUS or others	AUS; sling; other; PFMT

- **Counselling in every stratum:** expect **one thin pad/day** with either. AUS — check physical and cognitive ability; **lifespan 8–10 years;** replaceable. Sling — **failures mostly go to AUS.**
- **After surgery:** AUS failure → Fig. 123.12; sling failure → Fig. 123.13; other failure → re-evaluate; success → interval follow-up.
- **What it tests: radiation alone puts a man in the AUS-first stratum.** AUS lifespan is **8–10 years here but 7–10 years in the text** (guideline figure versus authors' estimate).

KEY POINTS — INDICATIONS

- Surgery for intrinsic sphincter deficiency **failing conservative care.**

- **Implant choice:** severity, prior surgery, bladder function, cystoscopy, dexterity, cognition, long-term reoperation risk, preference.
- **Slings only very selectively after radiation or for severe gravitational UI.**
- **No AUS with poor dexterity, cognitive disability, poor urethral tissue or severely reduced compliance.**

Concept — *The AUS works regardless of sphincter or detrusor function but demands a competent operator and healthy urethra. The sling demands a residual sphincter, an unirradiated urethra and a contractile bladder. Each contraindication is the mirror of the other device's requirement.*

SURGICAL DECISION MAKING IN THE HIGH-RISK PATIENT

- "Fragile urethra" = increased risk of AUS cuff erosion. Established risk factors: **prior pelvic radiotherapy, previous failed AUS (erosion or infection), previous urethroplasty, hypogonadism.**

Prior Pelvic Radiotherapy

- **Microvascular injury to the spongiosum**, despite modern delivery: erosion **about 3–33% irradiated versus 1–13% not**; irradiated men **erode sooner** (median about **1 versus 3 years**).
- **Radiation predicts the first explant, not failure of later implants** — those are driven by prior erosion and urethroplasty.

Previous Erosion/Infection

- **Coronary artery disease (CAD) and age >80** associated with erosion; **diabetes and smoking not shown independently** (small series — still optimise).
- **Prior erosion:** 5-year repeat erosion approaches **50%**.

Previous Urethroplasty

- **Greater risk than radiation** (multi-institutional); **10-year revision-free below 50%**.
- **Transection divides the bulbourethral arteries** (dual blood supply lost); benefit of **non-transecting urethroplasty is unproven.**

Low Testosterone

- **Independently predicts erosion** in small series (androgens support urethral health). **Androgen deprivation therapy not shown to raise risk; testosterone supplementation not shown to lower it.**

Strategies to reduce erosion in the high-risk patient

- **Medical optimisation. Erosion site disputed:** mostly ventral or lateral in one series; **equal dorsal, ventral, lateral** in a multi-institutional study.
- **Transcorporal (TC) AUS (George Webster, 2002): tunica albuginea interposed between dorsal urethra and cuff**, preserving vascularity and bulking the **thinnest (dorsal) spongiosum**; erosion benefit **mixed**.
- **PRB 51–60 cm H₂O or cuff 0.5 cm larger than measured** — scarce data. Many men still prefer **repeated AUS attempts** to pads, catheter or diversion.

EROSION RISK FACTORS — STRENGTH OF ASSOCIATION

Established / strong	Associated in some series	Mixed or not shown independently
Prior radiotherapy; prior erosion or infection; previous urethroplasty; urethral catheterisation (with or without deactivation); 3.5-cm cuff after radiation; tandem distal cuff	CAD; age >80; low testosterone; obesity; hypertension	Diabetes and smoking (not shown independently in high-risk data, yet listed as established in the erosion section); hypogonadism and prior revisions — mixed

- **What the table tests:** the chapter **contradicts itself on diabetes, smoking and hypogonadism**. Examinable safe answer: **radiation, prior erosion and urethroplasty are the strong three**; the others are debated.

KEY POINTS — HIGH-RISK PATIENT

- The four risk factors above; **optimise medically; consider TC cuff and reduced-pressure PRB**.

Concept — Erosion is a **problem of spongiosal blood supply under constant pressure**. Every risk factor (radiation, transection, prior erosion, low androgen) reduces perfusion, and every countermeasure (TC cuff, lower PRB pressure, larger cuff) either adds tissue or lowers the pressure.

TECHNIQUE OF IMPLANTATION OF CURRENTLY APPROVED AND AVAILABLE DEVICES**Operative Preparation for All Implant Surgeries**

- **General or spinal; low lithotomy; shave; chlorhexidine–alcohol preparation** of lower abdomen, genitals, perineum.
- **Intravenous antibiotics within 60 minutes of incision** (AUA Best Practice Statement): **aminoglycoside 5 mg/kg + first- or second-generation cephalosporin or vancomycin**.
- **Chlorhexidine scrub + iodine povacrylex–alcohol (DuraPrep)** cut skin culture rates versus povidone-iodine, but **not implant infection**. Optional urethral catheter aids identification of the urethra.

Artificial Urinary Sphincter**Bulbar Cuff Artificial Urinary Sphincter**

- **Site:** proximal bulbar urethra, perineal incision, proximal to corporal fusion — **safe circumferential dissection, cuff protected from opening on sitting, largest spongiosal diameter**.

Sequence: incise skin → Colles fascia → bulbospongiosus → **incise Buck fascia** where it reflects off the urethra onto the diverging corpora → create a **2-cm tunnel (cuff width) under vision, dorsal to Buck fascia** over the urethral roof → pass a right-angle clamp → **measure the circumference around the spongiosum** → choose cuff (**usually 4.0 or 4.5 cm**) → place → pass tubing through bulbospongiosus into the deep perineal space beneath Colles fascia.

- **Sizing:** a **consistent method** matters more than catheter in or out. **Cuff is slightly tighter than the tape**. Clearly wrong → next size. **Less tight cuffs give better continence**. Best test = **visual and endoscopic appearance**.
- **Sizes 3.5–11 cm** (0.5-cm steps to 7.5, then 8, 9, 10, 11); **large sizes for bladder neck**.
- **3.5-cm cuff:** for atrophy after previous AUS or very distal placement; raises erosion after radiation → prefer TC and/or 51–60 cm H₂O PRB.

Fig. 123.6 — (A) Ideal first-time cuff site, **proximal to convergence of the corpora cavernosa**. (B–D) Bulbospongiosus divided; dissection at the **spongiosum–crus interface on each side**.

Pressure-Regulating Balloon

- **Scrotal, perineal or abdominal**, by incisions, habitus, preference; know all three.
- **Abdominal**: lower-quadrant incision **ipsilateral to pump** → rectus or external oblique fascia → split muscle → preperitoneal space → digital pocket → close fascia round tubing.
- **External ring** (penile-implant technique, subinguinal or perineal incision): perforate inguinal floor into retropubic space; or **ectopic submuscular** above transversalis fascia — **no sutures**, fascia prevents extrusion. **Contraindicated after mesh hernia repair, cystectomy or extensive abdominal surgery** → abdominal route avoids bladder or bowel injury.
- **Fill 23 mL** (more for bladder neck): saline, or **contrast diluted with sterile water for isotonicity** (Cysto-Conray 60 + 15 mL; Hypaque 25% 50 + 60 mL). **No contrast with iodine allergy.**

PRB PRESSURE SELECTION

PRB (cm H ₂ O)	Use
51–60	High-risk bulbar (radiation, prior erosion, atrophy) — trades continence for erosion protection
61–70	Standard bulbar cuff
71–80	Bladder neck cuff

- **Trap**: treating atrophy by **raising PRB pressure increases erosion** (see Urethral Atrophy). Pressure goes **down** for risk and **up** only for the bladder neck.

Control Pump Placement

- **Anterior scrotum, subdartos**. After abdominal PRB: dissect deep to Scarpa fascia → **invaginate scrotal skin with a finger** → strip fascia off dartos → pocket (Fig. 123.8). Single incision: pump points **footward** (scrotal) or **headward** (perineal).

Making the Connections

- **In the abdominal incision when used** (less wear, easy re-exploration); **quick connectors**.
- **Cycle; urethroscopy: closed cuff → slight blanching; open cuff → minimal coaptation**; excessive coaptation when open → **larger cuff** (Fig. 123.7).

Tandem Cuff Artificial Urinary Sphincter

- **Up to 11% leak significantly after one cuff**; for **incomplete occlusion**, a second cuff helps. **Not first-line: distal cuff erodes more and tandem uses up revision sites.**

Sequence: perineal incision → dissect the distal bulbar urethra **at least 1.5–2 cm distal to the primary cuff** → place new cuff → divide existing cuff tubing → join with a **metal Y-connector** → **add 3 mL fluid** → **leave deactivated for 6 weeks**.

Transcorporal Cuff

- **Indications**: difficult first or revision surgery; urethra **<3.5 cm**. After prior erosion or urethroplasty, place the TC cuff **at a different site** along the bulbar urethra.

Sequence: **parallel longitudinal incisions in the tunica albuginea** of each corpus near the spongiosal interface → dissect **through the intercorporal septum** from side to side → the **ventral tunica stays attached to the dorsal spongiosum** → **imbricate the lateral tunical edges behind the urethra** (reduces haematoma) → measure → place cuff (Figs. 123.9D, 123.10).

- **Erections usually preserved** in sexually active men despite theoretical risk.
- **Advantages**: **less urethral injury at reoperation** (prior erosion, injury, urethroplasty, rectourethral fistula repair); **better fit** from added bulk; **lower erosion unproven**. Ideal when atrophy leaves a **small distal urethra (≤3.5-cm cuff)** — a 3.5-cm cuff, prior radiation and prior erosion each raise erosion risk.

- **Downside:** more transient retention needing suprapubic cystostomy. Early high-risk series: **76–84% cured or improved, no erosion**; overall evidence mixed (series mix standard and TC).
- **Variants:** "Gullwing" — wider incisions, **complete tunical wrap** (circumferential, not just dorsal, protection; no long-term data). **Cuff around bulbospongiosus** protects the ventral urethra — erosion very rare, revisions high.

Transscrotal Artificial Urinary Sphincter

- **Wilson technique:** all components via **one transverse scrotal incision**; designed for reoperation with atrophy (distal tandem or TC cuff). **Inferior continence** — **cannot reach the wide proximal bulb**. Enhanced-exposure modification allows larger proximal cuffs.

Perineal cuff	Transscrotal cuff
Proximal bulb, larger diameter (typically 4.5 cm); distal sites kept for revision; authors' preference	More distal, smaller urethra (typically 4.0 cm); single incision; inferior continence unless modified exposure is used

Bladder Neck Artificial Urinary Sphincter

- **Optional, more invasive, harder; prostate must be intact** — e.g. myelomeningocele and neuropathic disorders.
- **Open:** lower abdominal incision, circumferential dissection at the vesicoprostatic junction. **Robotic:** bladder–seminal vesicle plane under vision, no laparotomy. Fill volume per manufacturer; **children** need different cuff and PRB.

Bulbar AUS	Bladder neck AUS
Standard; post-RP; cuff usually 4.0–4.5 cm ; PRB 61–70 ; perineal approach	Prostate intact, neuropathic; contraindicated after RP ; cuff ≥8 cm ; PRB 71–80 ; abdominal or robotic; less erosion and atrophy

KEY POINTS — IMPLANTATION

- Bulbar cuff **proximal to corporal convergence**; **PRB 61–70** (abdominal, subinguinal or perineal).
- **Deactivate for 4–6 weeks after surgery** — essential to heal without erosion.
- Non-mechanical revision may need **TC, 3.5-cm or tandem cuff**.

Concept — *Implantation is a trade between urethral bulk and future options. Go as proximal as possible, where the spongiosum is thickest. Keep distal sites in reserve. When bulk is lacking, borrow it (TC tunica) rather than squeeze harder (higher PRB or smaller cuff).*

MALE SLINGS

Transobturator Bulbourethral Sling

- **Rehder and Gozzi (2007).** For SUI after CaP treatment, TURP and laser enucleation. **Advance XP** (better anchoring) replaced Advance.

Sequence: perineal incision → **open bulbospongiosus**, free it from spongiosum to the **perineal body** → mark, then divide perineal body 1–2 cm to **release the proximal bulb** → open spongiosum–corpus space until **inferior pubic ramus palpable behind each crus** → stab incision **one fingerbreadth below adductor longus** in groin crease; spinal needle finds foramen → **helical trocar outside-in (two "pops")**, exiting onto the fingertip at the **spongiosum–ramus apex** → draw mesh out **without tension**; proximal edge at marked perineal body → **four absorbable sutures to spongiosum** → **tension arms equally under cystoscopy until the bulbar mucosa coapts** (bulb moves **3–4 cm proximally and cephalad**; Fig. 123.11) → remove sheaths, trim and bury arms → close in layers.

Quadratic Sling

- **Virtue:** **prepubic fixation matters**; durable at 3 years in mild UI. **Role undefined**, but compression may suit wider severity and causes beyond RP.

Adjustable Continence Devices

ProACT: contrast-filled bladder on fluoroscopy → **sharp trocar via perineum** → past pelvic diaphragm **swap to blunt trocar** → bladder neck → deploy balloon via U-channel sheath, fill with saline → repeat other side → **scrotal port tunnels** → close, remove catheter.

ATOMS: perineal incision, **bulbospongiosus exposed, not opened** → needles through **superomedial foramen**, round the ischiopubic ramus, out through the **triangle of bulbospongiosus (medial), ramus (lateral), transverse perineal muscle (posterior)** → sutures pull the arms through → **cushion midline under bulb**, arms fixed, mesh trimmed → **subdartos scrotal port, 9 mL saline**.

Advance XP	ATOMS
Bulbospongiosus opened ; perineal body released; tensioned at surgery ; no port; repositioning	Bulbospongiosus left intact ; cushion under bulb; adjusted later via scrotal port (9 mL); compression

KEY POINTS — MALE SLINGS

- Slings need **mild-to-moderate SUI, contractile bladder, well-vascularised non-irradiated urethra**; transobturator slings **depend on an intact external sphincter**.
- Quadratic and adjustable devices: **limited efficacy and durability data**.

Concept — The Advance XP is **tensioned once, at surgery, under endoscopic view**, so its success depends on the sphincter it lifts. Adjustable devices move the **titration into the clinic**, at the cost of a port and more complications.

EVALUATION OF PERSISTENT INCONTINENCE AFTER ARTIFICIAL URINARY SPHINCTER AND SLING

Incontinence After Previous Artificial Urinary Sphincter

- **Differential** (tabulated) plus rare **plugged delay-fill resistor or kinked tubing**; insufficient compression includes **cuff oversizing**. General work-up as for de novo SUI (Fig. 123.3).

Cause	Clue	Confirm
Deactivation	Sudden loss	Device cycles actively → excluded. If deactivated with too little fluid to cycle: squeeze the pump's lateral edges, press opposite the deactivation button with a cotton bud, or hold the button ≥30 seconds
Mechanical failure / fluid leak	Sudden loss	PRB imaging during cycling — plain film (contrast fill) or ultrasound (saline fill): PRB loses volume
Atrophy	Slow, progressive worsening	PRB changes size with cycling and refills passively → mechanical failure unlikely. Cystoscopy shows incomplete coaptation during cycling
Erosion	—	Cystoscopy
Storage failure / DO	Urgency	UDS
PRB herniation	—	Exclude — the PRB must sit deep pelvic or submuscular

- **Isolated PRB replacement** proposed for intact system without atrophy (intrinsic PRB failure).

FIG. 123.12 AUA/SUFU ALGORITHM FOR ARTIFICIAL URINARY SPHINCTER FAILURE

From Sandhu JS, Breyer B, Comiter C, et al. Incontinence after prostate treatment: AUA/SUFU Guideline. J Urol. 2019;202(2):369–378.

Presentation	Action
Infection	Explant urgently; possible replacement 3–6 months later
Erosion	Explant urgently; urethral catheter about 3 weeks; possible replacement 3–6 months later
Worsening UI, no infection or erosion → cross-sectional imaging of PRB volume	PRB with little or no fluid → replace whole AUS or the failed parts . PRB with normal fluid (>20 mL) → downsize cuff, move cuff, add cuff, or increase PRB pressure

- **Note:** the algorithm lists "increase PRB pressure", but the chapter warns this **raises erosion risk**. Prefer the cuff options.

Incontinence After Previous Sling

- **Know the sling type** (Advance, Virtue, bone-anchored) and **timing**.
- **Advance loosened in first 6 weeks** (improved, then recurred) → **re-explore**: cut arms lateral to urethra → remove central portion from spongiosum (extensive if much elevation) → new sling.
- **Persistent or delayed SUI** → **DO or SUI? Sling failure** → **AUS is gold standard**, results as for first-time AUS. **Cystoscopy in every worsening sling patient** (mesh erosion).
- **Second Advance for late failure** (slippage): about **56% success**; only if **AUS contraindicated or refused**.

FIG. 123.13 AUA/SUFU ALGORITHM FOR SLING FAILURE

From Sandhu JS, Breyer B, Comiter C, et al. Incontinence after prostate treatment: AUA/SUFU Guideline. J Urol. 2019;202(2):369–378.

Presentation	Action
Infection or erosion	Explant as much as possible; consider AUS 3–6 months later
Inadequate continence	Place AUS (with or without sling explant), or replace the sling — only by providers trained in replacement

KEY POINTS — PERSISTENT INCONTINENCE

- Distinguish the five causes; **cystoscopy** excludes erosion and shows cuff coaptation (atrophy).

Concept — Tempo of failure points to the cause: sudden means the device (deactivation or leak), gradual means the urethra (atrophy). PRB volume then separates the two before anyone operates.

ARTIFICIAL URINARY SPHINCTER COMPLICATIONS**Urinary Retention**

- Rare; commoner with TC and 3.5-cm cuffs.

Sequence: confirm cuff deactivated → small urethral catheter for 24–48 hours → fails a second trial at 48 hours → **suprapubic cystostomy** (to cut erosion risk), under **ultrasound or fluoroscopic guidance** so the PRB is not punctured or contaminated → retention lasting several weeks → **cuff undersized** → reoperate and replace the cuff; review preoperative urodynamics.

- **Late-onset retention** → endoscopy and UDS to exclude **proximal obstruction, erosion or detrusor failure**.

Artificial Urinary Sphincter Infection

- **1–3% at first implant; up to 10% high-risk.** Skin flora — *Staphylococcus epidermidis*, *S. aureus*. **InhibiZone** (rifampicin + minocycline) **has not cut AUS infection** as it did for penile implants.
- **Late (>4 months):** indolent inoculum or **haematogenous spread** → **prophylaxis before urinary tract manipulation**.
- **Scrotal pain is the key early sign** ($\pm$ erythema, oedema, pus). **Antibiotics fail** → **almost always explant**; traditionally reimplant months later.
- **Immediate salvage (non-eroded):** **remove all** → **seven-solution antiseptic irrigation** (penile-prosthesis protocol) → **new device at once**; successful in most.
- **Contraindications to salvage:** sepsis, ketoacidosis, necrotising infection, device erosion, immunosuppression, gross pus at explant.

Urethral Erosion

- **2–15%; up to 35% with multiple risk factors** (see risk table). **Delayed activation lowered erosion**, especially at reimplantation.
- **Remove all components** — **assume infected**, though most left-in PRBs stay clean. **Wound treated as infected:** loose closure or secondary intention.
- **Stricture after erosion: 8–35%; prior radiation, not erosion extent**, predicted it in one series.

MANAGEMENT OF CUFF EROSION COMPARED

Strategy	What is done	Bottom line
Catheter only	Urethral catheter; urethrogram confirms healing before removal	Traditional; adequate for <50% urethral loss ; higher stricture risk if >50% loss
Pseudocapsule closure (urethrorrhaphy)	Close fibrous capsule	Did not prevent stricture
In situ repair without mobilisation	Ventral mucosal approximation	Fewer strictures than catheter alone
Mobilisation + spatulated anastomotic urethroplasty	Formal repair	Better than catheter for >50% loss in one series; no significant difference in a multi-institutional study

- **Rule: <50% loss → catheter; >50% loss → optimal approach unresolved.** Match aggressiveness to stricture risk, weighed against contamination and the **harm of mobilising in an infected field**. **Complete cuff erosion carries the highest stricture risk.**
- **Reimplant at 3–6 months after confirming patency** (cystoscopy or retrograde urethrogram), **proximal or distal to the old site** (scar, poor supply).
- **Repeat erosion up to 35% in the first months; nightly deactivation** advised after reimplantation for erosion; **TC and/or low-pressure PRB** may help.

Urethral Atrophy

- **Chronic compression of spongy tissue; insidious**, unlike device failure; **a commonest cause of revision**.
- **Options: downsize; relocate to thicker spongiosum; TC; tandem.** **Never simply raise PRB pressure** — more erosion.
- **Authors: downsize in place** → if already **4.0 cm**, move **proximally**, or **distally with TC**. In revision series most cuffs stayed in place (distal moves outnumber proximal) and downsizing predominated, with **marked fall in pad use**.

- **Atrophy disputed:** one centre treats all failures as mechanical, **same-size cuff**, acceptable results; pressure profilometry finds true atrophy in a **minority**.

Mechanical Failure

- **Device life 7–10 years. Within 3 years** (no infection or erosion): **replace the failed part** — usually **pump or cuff**, median ≤ 1 year.
- **>3 years: PRB is the usual culprit** → **replace entire device**; slow PRB leaks hide at surgery. **Ohmmeter or burette testing** of the PRB may spare total replacement.

KEY POINTS — AUS COMPLICATIONS

- **Second failed trial at 48 hours** → **suprapubic cystostomy**. Erosion management graded to severity and contamination. **TC $\pm$ low-pressure PRB after erosion. Replace whole AUS if >3 years old.**

Concept — Every complication is managed to **protect the urethra for the next device**. *Suprapubic rather than urethral drainage, total explant at erosion, a new cuff site, and whole-device replacement after 3 years all serve the same aim: a salvageable urethra.*

SPECIAL CIRCUMSTANCES

Managing Device Implantation With Spongiosal Vascular Insufficiency

- **Protect spongiosal blood supply:** **non-transecting urethroplasty** for strictures before AUS; **TC AUS, lower-pressure PRB**; possibly **testosterone replacement** if **hypogonadal** (unproven).

Urethral and Vesicourethral Anastomotic Stricture

- **Stricture at the cuff** (compression or ischaemia) **may herald erosion**; periodic dilation has worked. **Proximal to the cuff** → **balloon dilation under vision** or **holmium laser** via flexible scope.
- **ROBUST III RCT** — U.S. Food and Drug Administration (FDA)-approved **paclitaxel-coated urethral balloon** versus direct vision internal urethrotomy (DVIU) or dilation in recurrent bulbar stricture: **1-year freedom from intervention 83.2% versus 21.7%; 67% functional success at 3 years. No data yet for VUAS.**
- **Post-erosion stricture** → **urethroplasty** → **repeat AUS** (often TC): **about a quarter re-erode within 2 years.**
- **VUAS after AUS** → **balloon or ureterscopic laser. Large scope (22–24 Fr) proximal to cuff** → **surgically uncouple the cuff. Refractory** → **robotic reconstruction**, sparing the cuff.
- **Timing:** traditionally treat VUAS **before AUS**; but **non-obiterated VUAS (<12 Fr) dilated synchronously** showed **no difference** in retreatment or erosion.

Urinary Incontinence After Cystectomy and Neobladder Reconstruction

- **SUI after orthotopic neobladder: 33–95%**; continence drives QoL.
- **AUS helps about three-quarters, but 60% need revision (about 2 years)**, over a third for erosion or infection — **radiotherapy, bacterial colonisation, clean intermittent catheterisation (CIC).**
- **PRB ectopic, lateral counter-incision, behind rectus** — avoids neobladder or bowel.

Dual Implantation of Artificial Urinary Sphincter and Inflatable Penile Prosthesis

- **Synchronous = metachronous AUS + inflatable penile prosthesis (IPP)** for device survival and reoperation (meta-analysis); high satisfaction. **Dual implants are revised more than either alone** (about 2 $\times$ IPP, 1.7 $\times$ AUS).

- **Synchronous: penoscrotal (IPP) + perineal (AUS). Metachronous: AUS first**, IPP after activation — continence is usually wanted first.
- **Existing IPP + TC cuff**: dissect **between tunica and pseudocapsule**; close corporotomy with **allograft** (e.g. Tutoplast). **Neobladder**: dual implants as safe as in controls.

Climacturia

- **20–68% after RP**, including **leakage on arousal**. Condom and pre-coital voiding help little; authors use sling or AUS by severity.
- **Mini-jupette at IPP**: **3-cm corporotomies** → **graft sutured to medial edges** → **inflation tightens graft across urethra**; **partial inflation** treats mild SUI. 5-year data: about 90% improved, **no erosion or infection**.

KEY POINTS — SPECIAL CIRCUMSTANCES

- VUAS after AUS → **balloon or ureteroscopic laser**; **synchronous AUS + IPP safe**; **mini-jupette** for erectile dysfunction with climacturia and mild SUI.

Concept — *In each special case the **cuff is the fixed asset**. Instrumentation is minimised, the cuff is uncoupled or bypassed, and other devices are routed away from it, because anything that stresses the bulbar urethra risks losing continence.*

SLING COMPLICATIONS

COMPLICATIONS BY DEVICE CLASS

Device	Commonest complication	Erosion / explant	Other
Nonadjustable (transobturator)	Pain and wound problems ; groin pain from trocar is self-limiting (median visual analogue scale (VAS) 0 at 3 months)	Erosion or explant 0–0.4% — exceedingly rare	Chronic pain 1.3%; urgency or retention 4–10%
ATOMS (adjustable)	Complications more frequent — limits adoption	Erosion and infection about 10%; explant 10–16%	Chronic pain 1.5%
ProACT (non-circumferential)	Bladder and/or urethral perforation (about 5%)	Erosion about 4%	Leak about 4%; migration about 6.5%; revision 22%

- **Pattern: adjustability costs complications**. The fixed transobturator sling almost never erodes; the port-bearing adjustable devices erode, migrate and need revision. Trap: **perforation, not erosion, is the commonest ProACT complication**.

Concept — *The **non-compressive transobturator sling barely touches urethral blood supply**, so it rarely erodes. Every device that compresses — adjustable sling, balloons, AUS — carries erosion as its signature risk.*

GAPS AND FUTURE DIRECTIONS

Innovations and Emerging Concepts in Devices

- **AUS: repeated revisions and morbidity**. Slings and compression devices: frequent complications, often short of goals. Future: **electronic control, patient-specific dynamic pressure, new devices**.

Registries

- **Long-term data are small, retrospective, single-institution** → registries and prospective trials (AUS versus sling, ATOMS versus AUS, Virtue).

Surgical Training

- **>150,000 men** implanted, yet **many urologists never do SUI surgery**; **fellowships and simulation** should widen access.

SUMMARY

- **Demand persists** with ongoing surgery for localised CaP. Work-up adds **pad test, uroflowmetry, PVR**; cystoscopy and UDS selectively.
- **AUS = preferred for moderate-to-severe and complex SUI. Slings** for mild-to-moderate UI **without prior urethral surgery, radiotherapy or detrusor acontractility**; unproven for neurogenic or post-traumatic UI.
- **AUS-sling comparison is confounded** by severity, endpoints, follow-up and **selection bias**; MASTER is the only RCT.
- **Nearly 90% improve in urinary QoL, whatever the severity.** Erosion and mechanical failure drive innovation.

Concept — *The chapter reduces to one decision: **how much sphincter, how healthy a urethra and how capable a patient.** A residual sphincter and good tissue allow a sling; anything less, or radiation, points to the AUS. Everything after implantation is about preserving the urethra for the next revision.*